

B. Tech. I (CSE (SEM-I))
MST-II Applied Physics

Max. Marks:15

Time: 1 Hour

- Q1** a). What is Rayleigh's Criteria of resolution?
b). What is negative crystal? Give example.
c). What factors are causing attenuation in optical fiber?
d). Mention the applications of Laser.
e). Write the value of refractive index for Canada balsam material used in Nicole prism. (5×1)
- Q2.** a). Distinguish between step index and graded index optical fiber.
b). Calculate numerical aperture and acceptance angle of an optical fiber having refractive indices of core and cladding 1.60 and 1.50, respectively. (3,2)
- Q3.** Discuss the principle, construction and working of four level laser. or
What are the classification of diffraction? Explain the maximum and minimum conditions of fraunhoffer diffraction at double slit? (5)

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Q1. ✓ (a) Mention the applications of optical fiber.

(b) Define metastable states?

(c) What are factors causing attenuation of signal in optical fiber?

(d) What is a diffraction grating?

(e) Expand the term LASER?

(5×1)

Q2. ✓ (a) Calculate numerical aperture and acceptance angle of a fiber having refractive indices of core and cladding 1.50 and 1.45, respectively.

(b) Distinguish between step index and graded index optical fibers.

(2,3)

Q3. Explain spontaneous and stimulated emission of light. Discuss the Einstein's coefficients and find relation between these coefficients?

or

✓ Mention two types of diffraction. Discuss Fraunhofer diffraction at circular aperture.

(5)

Department of Basic and Applied Sciences, Punjabi University, Patiala.

B.Tech.-I

MST-II

Applied Physics-I

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Q1. (a) Why cooling is required in Ruby laser?

(b) What are the causes of signal attenuation in optical fibre communication?

(c) What is numerical aperture of an optical fiber?

(d) What is a de-Broglie wave?

(e) What do you understand by tunneling effect?

(1×5=5)

Q2. Derive an expression for energy of a particle confined to move in one dimensional box of length L. **(2)**

Q3. Calculate the numerical aperture, acceptance angle and the critical angle of a fiber having core refractive index= 1.50 and cladding refractive index=1.45. **(3)**

Q4. What is stimulated emission? Derive the Einstein's relations describing the probabilities of stimulated absorption and stimulated emission.

OR

What do you mean by a wave function? Starting with the time dependent wave equation derive an expression for the time independent Schrodinger's equation. **(5)**